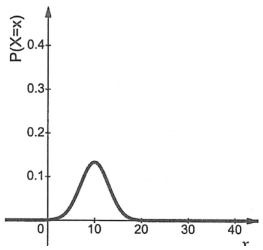
The Normal Distribution

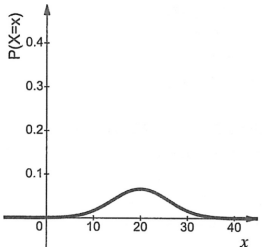
Questions 1–65

Question 1. The four diagrams below show $X \sim \mathrm{N}(10, 1)$, $X \sim \mathrm{N}(10, 3)$, $X \sim \mathrm{N}(20, 3)$ and $X \sim \mathrm{N}(20, 6)$. Which is which?

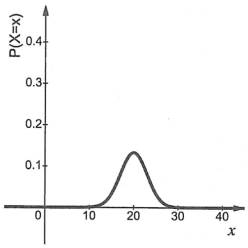
A



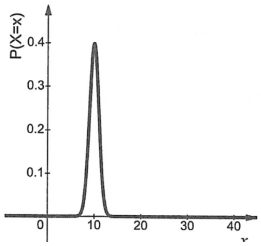
B



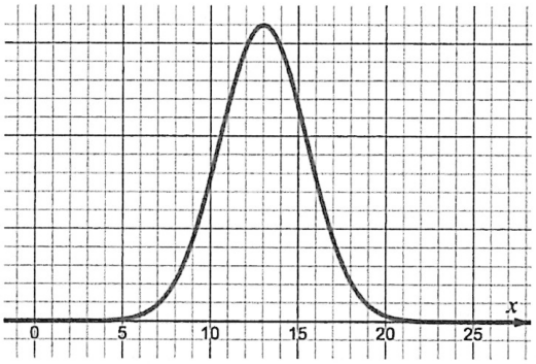
C



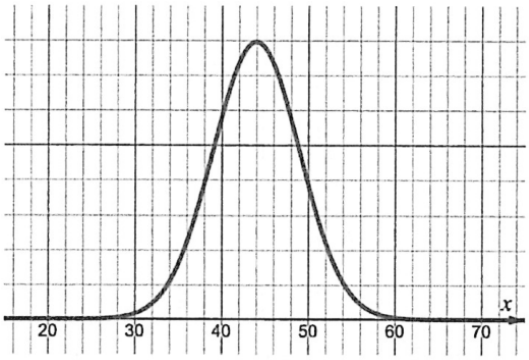
D



Question 2. It is given that the distribution shown below is normal. Use the diagram to estimate the standard deviation.



Question 3. It is given that the distribution shown below is normal. Use the diagram to estimate the standard deviation.



Question 4. Find the standard deviation of each of these normal distributions:

a) $X \sim \mathrm{N}(72, 49)$
b) $X \sim \mathrm{N}(12, 5)$
c) $X \sim \mathrm{N}(53, \sqrt{2})$

Question 5. Approximately 99.73% of the area under a normal distribution is within three standard deviations of the mean. For each of these normal distributions find the values of a and b such that $\mathrm{P}(a < X < b) \approx 0.9973$:

a) $X \sim \mathrm{N}(30, 25)$
b) $X \sim \mathrm{N}(18, 12.25)$
c) $X \sim \mathrm{N}(605, 1024)$

Question 6. Sketch the following normal distributions. You do not need to consider their heights.

a) $X \sim \mathrm{N}(25, 5^2)$
b) $X \sim \mathrm{N}(100, 100)$
c) $X \sim \mathrm{N}(50, 5)$

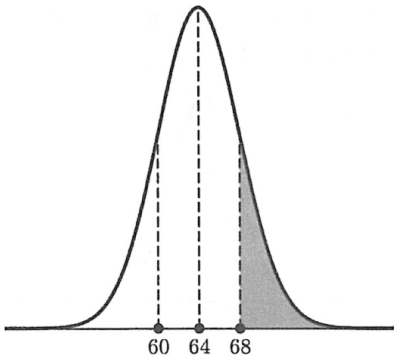
Question 7. Sketch $X \sim \mathrm{N}(60, 5^2)$ and $Y \sim \mathrm{N}(100, 5^2)$ on the same axes, identifying which is which.

Question 8. Sketch $X \sim \mathrm{N}(20, 5^2)$ and $Y \sim \mathrm{N}(30, 5^2)$ on the same axes, identifying which is which.

Question 9. Sketch $X \sim \mathrm{N}(60, 10^2)$ and $Y \sim \mathrm{N}(60, 5^2)$ on the same axes, identifying which is which.

Question 10. Sketch $X \sim \mathrm{N}(8, 1)$ and $Y \sim \mathrm{N}(10, 4)$ on the same axes, identifying which is which.

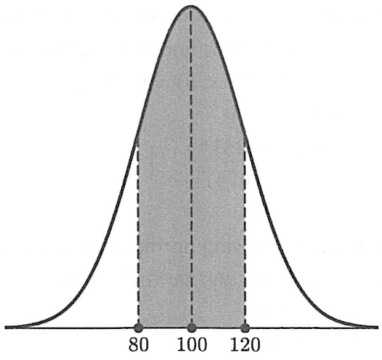
Question 11. It is given that $X \sim \mathrm{N}(64, 4^2)$. The graph of X is shown below and it is given that $\mathrm{P}(X > 68) = 0.158655$



Use the symmetry of the graph to find:

a) $\mathrm{P}(X < 68)$
b) $\mathrm{P}(X < 60)$
c) $\mathrm{P}(60 < X < 68)$

Question 12. It is given that $X \sim \mathrm{N}(100, 400)$. The graph of X is shown below and it is given that $\mathrm{P}(80 < X < 120) = 0.682690$



Use the symmetry of the graph to find:

a) $\mathrm{P}(X < 80)$
b) $\mathrm{P}(X > 120)$
c) $\mathrm{P}(80 < X < 100)$

Question 13. It is given that $X \sim \mathrm{N}(8, 4)$ and $Z \sim \mathrm{N}(0, 1)$. Find the corresponding values of Z when:

a) $X = 7$
b) $X = 9$
c) $X = 10.2$

Question 14. It is given that $X \sim \mathrm{N}(16, 16)$. By converting to the standard normal distribution, find:

a) $\mathrm{P}(X < 15)$
b) $\mathrm{P}(X > 17)$
c) $\mathrm{P}(13 < X < 15)$
d) $\mathrm{P}(18 < X < 19)$

Question 15. Given $X \sim \mathrm{N}(40, 8^2)$, use a calculator to find:

a) $\mathrm{P}(X \leq 37)$
b) $\mathrm{P}(X < 37)$
c) $\mathrm{P}(X \geq 45)$
d) $\mathrm{P}(X > 45)$

Question 16. Given $X \sim \mathrm{N}(800, 45^2)$, use a calculator to find:

a) $\mathrm{P}(X < 860)$
b) $\mathrm{P}(X > 810)$
c) $\mathrm{P}(780 < X < 850)$
d) $\mathrm{P}(860 < X < 880)$

Question 17. Given $X \sim \mathrm{N}(45, 3^2)$, find:

a) $\mathrm{P}(X = 46)$

b) $\mathrm{P}(X \neq 40)$

Question 18. Given $X \sim \mathrm{N}(80, 30)$, find:

a) $\mathrm{P}(X < 80)$

b) $\mathrm{P}(X < 75)$

c) $\mathrm{P}(X > 83)$

d) $\mathrm{P}(75 < X < 83)$

Question 19. Given $X \sim \mathrm{N}(125, 88)$, find:

a) $\mathrm{P}(X < 110)$

b) $\mathrm{P}(125 < X < 130)$

c) $\mathrm{P}(130 < X < 135)$

Question 20. Use the standard normal distribution, $Z \sim \mathrm{N}(0, 1)$ to find:

a) $\mathrm{P}(-1 < Z < 1)$

b) $\mathrm{P}(-2 < Z < 2)$

c) $\mathrm{P}(-3 < Z < 3)$

d) $\mathrm{P}(-4 < Z < 4)$

Question 21. Given $X \sim \mathrm{N}(200, 15^2)$ and your answers to the previous question, find:

a) $\mathrm{P}(185 < X < 215)$

b) $\mathrm{P}(170 < X < 230)$

c) $\mathrm{P}(155 < X < 245)$

d) $\mathrm{P}(140 < X < 260)$

Question 22. Given $X \sim \mathrm{N}(\mu, \frac{1}{4}\mu^2)$, find $\mathrm{P}(X < 0.9\mu)$

Question 23. Given $X \sim \mathrm{N}(5\mu, 16\mu^2)$, find $\mathrm{P}(X > \frac{11}{2}\mu)$

Question 24. Given $X \sim \mathrm{N}(2\mu, 81\mu^2)$, find $\mathrm{P}(0.1\mu < X < 0.5\mu)$

Question 25. 200 university students are surveyed and it is found that the monthly rent spent on student accommodation, £X, had the following results, where $\bar{x}$ denotes the sample mean:

$$\sum x = 105000 \text{ and } \sum (x - \bar{x})^2 = 440000$$

a) Find the sample mean and sample standard deviation of X
b) Using a normal distribution to model X, find the probability that a university student will pay more than £535 each month.

Question 26. 100 sixth form students are surveyed, and it is found that the number of hours of driving lessons, X, had the following results:

$$\sum x = 4650 \text{ and } \sum x^2 = 219000$$

a) Find the sample mean, $\bar{x}$, of X
b) It is given that the sample variance,

$$s^2 = \frac{\sum x^2}{n-1} - \frac{(\sum x)^2}{n(n-1)}$$

Find s^2
c) Assuming that X be modelled by $\mathrm{N}(\bar{x}, s^2)$, find $\mathrm{P}(43 < X < 46)$

Question 27. The heights of male NBA basketball players have a mean of 2.01 metres and a standard deviation of 10 cm. It is known that 99% of the basketball players are between 1.79 m and 2.23 m. Comment on whether it would be suitable to model the heights of male NBA basketball players using a normal distribution.

Question 28. Using the standard normal distribution, $Z \sim \mathrm{N}(0, 1)$,

a) Find a such that $\mathrm{P}(Z < a) = 0.25$
b) Find b such that $\mathrm{P}(Z < b) = 0.9$
c) Find c such that $\mathrm{P}(Z > c) = 0.01$
d) Find d such that $\mathrm{P}(Z > d) = 0.8$

Question 29. Given $X \sim \mathrm{N}(27, 4^2)$,

a) Find a such that $\mathrm{P}(X < a) = 0.02$

b) Find b such that $\mathrm{P}(X < b) = 0.7$

c) Find c such that $\mathrm{P}(X > c) = 0.21$

d) Find d such that $\mathrm{P}(X > d) = 0.6$

Question 30. It is given that $X \sim \mathrm{N}(800, 900)$ and $\mathrm{P}(a < X < b) = 0.9$. It is also given that $\mathrm{P}(a < X < 800) = \mathrm{P}(800 < X < b)$. Find the value of a and the value of b.

Question 31. It is given that $X \sim \mathrm{N}(65, 12)$ and $\mathrm{P}(64 < X < a) = 0.25$. Find the value of a

Question 32. It is given that $X \sim \mathrm{N}(18, 10)$.

a) Find a given that $\mathrm{P}(X > a) = 0.3$

b) Find b given that $\mathrm{P}(18 - b < X < 18 + b) = 0.3$

Question 33. It is given that $X \sim \mathrm{N}(\mu, 10^2)$. Find the value of μ in each of these cases:

a) $\mathrm{P}(X < 40) = 0.5$
b) $\mathrm{P}(X < 50) = 0.3$
c) $\mathrm{P}(X > 200) = 0.7$
d) $\mathrm{P}(X > 70) = 0.01$

Question 34. It is given that $X \sim \mathrm{N}(33, \sigma^2)$. Find the value of σ in each of these cases:

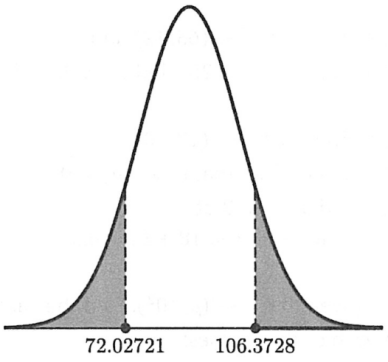
a) $\mathrm{P}(X < 30) = 0.2$
b) $\mathrm{P}(X < 50) = 0.6$
c) $\mathrm{P}(X > 1) = 0.99$
d) $\mathrm{P}(X > 34) = 0.49$

Question 35. The average monthly food bill, £X, for a family of 4 in the UK can be modelled by a normal distribution with mean £725. It is known that $\mathrm{P}(X < 700) = 0.35$. Find the standard deviation of X to one decimal place.

Question 36. Loose leaf tea is packaged in sachets. The mass, X, of loose leaf tea in each sachet is known to be normally distributed with a mean of 3 grams. It is given that 50% of the masses are between 2.6 g and 3.4 g. Determine the standard deviation of the masses.

Question 37. Category 2 ambulance calls are classified as an emergency that may require immediate assessment or on-scene intervention. Response times should have an average of 18 minutes and 90% must be responded to within 40 minutes. Explain why response times may not be normally distributed.

Question 38. The distribution $X \sim \mathrm{N}(\mu, \sigma^2)$ is shown below. The two shaded regions add to an area of 0.2.



a) Find μ and σ^2, given that the lower boundary is 72.02721 and the upper boundary is 106.3728
b) It is given that Y is the random variable such that $Y = 3X - 2$. Write down the distribution of Y.

Question 39. It is given that $X \sim \mathrm{N}(84, 5.29)$ can be transformed to $Y \sim \mathrm{N}(326.8, 72.4201)$ by the transformation $y = ax + b$. Find the values of a and b.

Question 40. It is given that $X \sim \mathrm{N}(\mu, \sigma^2)$. It is also given that $\mathrm{P}(X < 99) = 0.5$ and $\mathrm{P}(X > 90) = 0.6$. Find the value of μ and the value of σ

Question 41. It is given that $X \sim \mathrm{N}(\mu, \sigma^2)$. It is also given that $\mathrm{P}(X > 11) = 0.5$ and $\mathrm{P}(X > 12) = 0.05$. Find the value of μ and the value of σ

Question 42. It is given that $X \sim \mathrm{N}(\mu, \sigma^2)$. Find the value μ and σ in each of these cases:

a) $\mathrm{P}(X < 10) = 0.3$ and $\mathrm{P}(X < 11) = 0.35$
b) $\mathrm{P}(X < 50) = 0.4$ and $\mathrm{P}(X < 90) = 0.65$
c) $\mathrm{P}(X < 3) = 0.01$ and $\mathrm{P}(X > 9) = 0.02$
d) $\mathrm{P}(X > 67) = 0.44$ and $\mathrm{P}(X > 73) = 0.33$

Question 43. A banana is considered small if it weighs less than 140 grams, medium if it weighs between 140 and 215 grams, and large if it weighs over 215 grams. 14% of bananas are found to be small and 14% are found to be large. By writing down an appropriate distribution, estimate the weight of the lightest banana.

Question 44. Prerequisite knowledge for Q44–Q45: Points of Inflection, The Chain Rule & The Product Rule The normal distribution with mean 0 and standard deviation 1 has equation

$$y = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}x^2}$$

Use differentiation to find the x-coordinates of the points of inflection.

Question 45. The normal distribution with mean μ and standard deviation σ has equation

$$y = \frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Use differentiation to find the x-coordinates of the points of inflection.

Question 46. Prerequisite knowledge: Conditional Probability Given $X \sim \mathrm{N}(75, 10)$, find:

a) $\mathrm{P}(X < 70 \mid X < 80)$
b) $\mathrm{P}(X > 70 \mid X < 76)$
c) $\mathrm{P}(X < 77 \mid X > 73)$
d) $\mathrm{P}(X > 81 \mid X > 80)$

Question 47. A nursery grows yucca plants. When they are ready to be sold, their heights are normally distributed with a mean of 75 cm and standard deviation 5 cm. One yucca plant is selected at random. Find the probability that it is taller than 80 cm given that it is taller than 76 cm.

Question 48. The weight of sandbags is known to be normally distributed with a mean of 33 kg and standard deviation of 3.5 kg. A sandbag is selected at random. Given that it weighs more than 34 kg, find the probability that it weighs less than 36 kg.

Question 49. Prerequisite knowledge for Q49: The Binomial Distribution The weight, X, of a farm's eggs may be modelled by a normal distribution with mean 49 grams and variance 9 grams.

a) A single egg is randomly selected. Find the probability that the egg has a weight of less than 47 grams.

b) Eggs are put in boxes of twelve before they are transported to the local supermarket. A box of twelve eggs is randomly selected. Find the probability that at least four of the eggs has a weight of less than 47 grams.

Question 50. The lifetime, X, of a discount battery may be modelled by a normal distribution with mean 10 hours and standard deviation 1.2 hours.

a) A single discount battery is randomly selected. Find the probability that the battery has a lifetime:

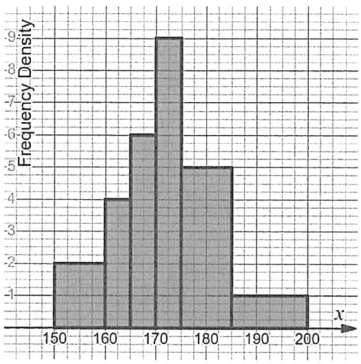
   i) equal to 11 hours;

   ii) greater than 11 hours.

b) Discount batteries are sold in packs of eight. A pack is randomly selected. Find the probability that at least two of the batteries have a lifetime of more than 11 hours.

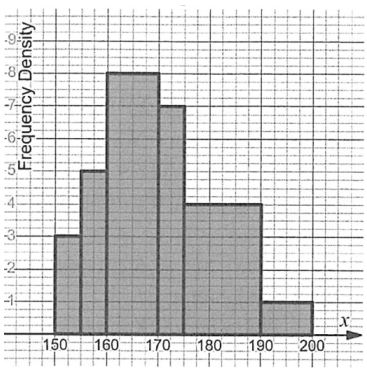
Question 51. The width, W, of a type of potato is known to be normally distributed with a mean of 5.7 cm and standard deviation 0.9 cm. A farmer is able to sell their potatoes to a large supermarket if $4.8 \leq W \leq 6.2$ for £0.25 per kilogram. If $W < 4.8$, the farmer sells the potatoes to a local farmer for £0.11 per kilogram. If $W > 6.2$, the farmer sells the potatoes to a local greengrocer for £0.21 per kilogram. Calculate how much the farmer will be able to sell a crop of 150 kg of potatoes for.

Question 52. The heights of 180 students were measured to the nearest cm and recorded in the histogram below.



a) Find $\mathrm{P}(x < 170)$
b) Find estimates for the mean and sample standard deviation of the heights.
c) Using your answers to b), model the data as a normal distribution, $Y \sim \mathrm{N}(\mu, \sigma^2)$. Find $\mathrm{P}(Y < 170)$ and thence comment on whether the normal distribution is an appropriate model for the data.

Question 53. The heights of a different group of 225 students were measured to the nearest cm and recorded in the histogram below.



a) Find $\mathrm{P}(x < 170)$
b) Find estimates for the mean and sample standard deviation of the heights.
c) Using your answers to b), model the data as a normal distribution, $Y \sim \mathrm{N}(\mu, \sigma^2)$. Find $\mathrm{P}(Y < 170)$ and thence comment on whether the normal distribution is an appropriate model for the data.

Question 54. Describe the conditions for which the binomial distribution $X \sim \mathrm{B}(n, p)$ can be approximated using the normal distribution $Y \sim \mathrm{N}(\mu, \sigma^2)$

Question 55. For each of the following binomial distributions, write down a normal distribution that could be used to approximate it, and comment on whether this would be appropriate.

a) $X \sim \mathrm{B}(500, 0.48)$
b) $X \sim \mathrm{B}(4, 0.5)$
c) $X \sim \mathrm{B}(150000, 0.001)$

Question 56. A circular spinner has five sections labelled 1, 2, 3, 4 and 5. The section labelled 2 is a sector with an angle of 160°. The spinner is spun 2000 times and the number, X, of times 2 is chosen by the spinner is recorded.

a) Write down the distribution of X and explain why it would be appropriate to approximate the distribution of X using a normal distribution.
b) Using a normal distribution to approximate the distribution of X, find a such that $\mathrm{P}(X > a) \approx 0.2$
c) Using the distribution of X and your value of a, find the actual value of $\mathrm{P}(X > a)$ to 3sf.

Question 57. It is given that $X \sim \mathrm{N}(300, 20^2)$ and $Y \sim \mathrm{N}(400, 30^2)$ where

$$\mathrm{P}(X > a) = \mathrm{P}(Y < a) = b$$

Find the value of a and the value of b

Question 58. It is given that $X \sim \mathrm{N}(16.5, 5.29)$ and $Y \sim \mathrm{N}(20.2, 6.76)$ where

$$\mathrm{P}(X > a) = \mathrm{P}(Y < a) = b$$

Find the value of a and the value of b

Question 59. It is given that $X \sim \mathrm{N}(50, 4)$ and $Y \sim \mathrm{N}(60, 9)$ where

$$\mathrm{P}(X < a) = \mathrm{P}(Y > a) = b$$

Find the value of a and the value of b

Question 60. It is given that $X \sim \mathrm{N}(200, 25)$ and $Y \sim \mathrm{N}(250, 49)$ where

$$\mathrm{P}(X < a - 15) = \mathrm{P}(Y > a + 15) = b$$

Find the value of a and the value of b

Question 61. Continuity Corrections The binomial distribution $X \sim \mathrm{B}(600, 0.52)$ is to be approximated using the normal distribution $Y \sim \mathrm{N}(312, 149.76)$. Approximate each of these probabilities.

a) $\mathrm{P}(X \leq 300)$
b) $\mathrm{P}(X > 320)$
c) $\mathrm{P}(X < 311)$
d) $\mathrm{P}(X \geq 295)$
e) $\mathrm{P}(X = 312)$

Question 62. Given that $X \sim \mathrm{B}(350, 0.6)$:

a) Find $\mathrm{P}(X \leq 200)$
b) Use a normal distribution, Y, to approximate your answer to a).
c) Find the percentage error between your answers to a) and b) to 3 significant figures.

Question 63. Given that $X \sim \mathrm{B}(540, 0.47)$:

a) Find $\mathrm{P}(X < 260)$

b) Use a normal distribution, Y, to approximate your answer to a).

c) Find the percentage error between your answers to a) and b) to 3 significant figures.

Question 64. Given that $X \sim \mathrm{B}(2000, 0.7)$:

a) Find $\mathrm{P}(X \geq 1440)$

b) Use a normal distribution, Y, to approximate your answer to a).

c) Find the percentage error between your answers to a) and b) to 3 significant figures.

Question 65. It is known that 4.5% of adults aged 18 and over use e-cigarettes.

a) In a random sample of 5000 adults, find the probability that:

i) More than 230 use e-cigarettes.

ii) More than 230 use e-cigarettes, given that less than 250 do.

b) Use a normal approximation to estimate your answers to a) i) and a) ii).